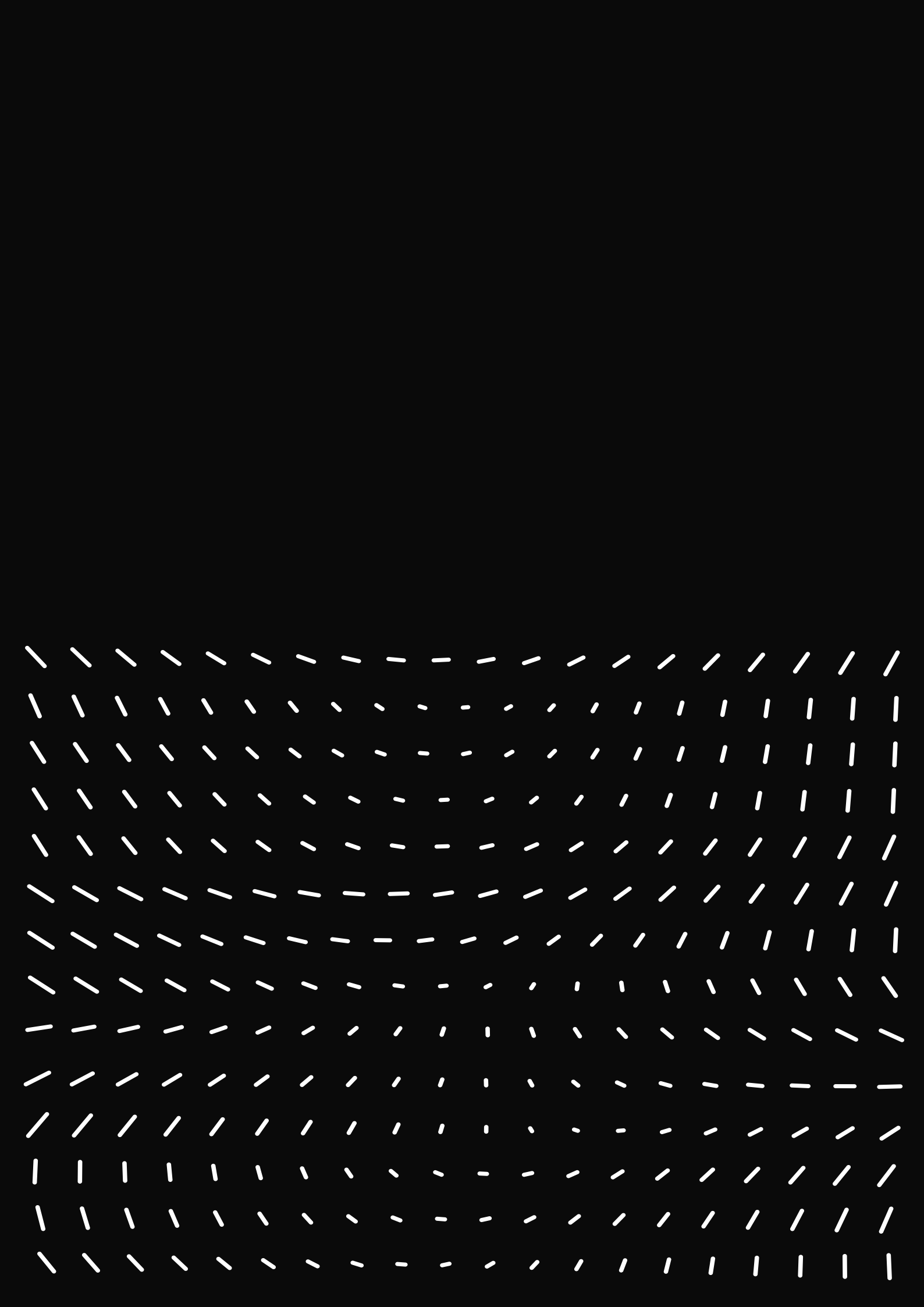




$$\int e^{-2x} dx$$

$$u = -2x \quad du = -2dx$$

$$\frac{1}{-2} \cdot du = dx$$

$$\int e^u \frac{1}{-2} du$$

$$-\frac{1}{2} e^u = -\frac{1}{2} e^{-2x}$$

$$\int x 2e^{-2x} dx \rightarrow 2 \int x e^{-2x} dx$$

$$\int u dv = uv - \int v du$$

$$u = x$$

$$du = dx$$

$$dv = e^{-2x}$$

$$v = \int e^{-2x} dx$$

$$u = -2x$$

$$du = -2dx$$

$$\frac{du}{-2} = dx$$

$$v = \int e^u \frac{du}{-2}$$

$$v = -\frac{1}{2} \int e^u du$$

$$v = -\frac{1}{2} e^u$$

$$v = -\frac{1}{2} e^{-2x}$$

$$\int u dv = uv - \int v du$$

$$x \cdot -\frac{1}{2} e^{-2x} - \int -\frac{1}{2} e^{-2x} dx$$

$$= \frac{-x e^{-2x}}{2} - \left(-\frac{1}{2} \int e^{-2x} dx \right)$$

$$= \frac{-x e^{-2x}}{2} - \left(-\frac{1}{2} \left(-\frac{1}{2} e^{-2x} \right) \right)$$

$$= \frac{-x e^{-2x}}{2} - \left(\frac{1}{4} e^{-2x} \right)$$

$$= \frac{-x e^{-2x}}{2} - \frac{e^{-2x}}{4}$$

$$f = \frac{-x}{2 e^{2x}} - \frac{1}{4 e^{2x}}$$

$$\int_1^2$$